

Experiment 2: PV Panel, Battery, and Direct Energy Transfer

ECEN 5517-012

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BENCH 5

I. Introduction

The goal of this experiment is to characterize the solar panel under various test conditions by making an IV curve and extrapolating key physical characteristics of the panel as a power generator. It will also characterize the battery under direct energy transfer.

II. Experimental Procedure and Results

a. Starting the Experiment

i. Powered the Cart using the Battery

Nameplate Data: 85 W power rating, 5.45 A short circuit current, 4.95 A rated current, 22.2V open circuit voltage, 17.2V rated voltage, 20 A series fuse, 600V max system open circuit voltage, Class C fire rating, and wiring is to be done with only 14 AWG copper wire or larger.

ii. Gathered Materials ✓

iii. Gathering Data About Energy Consumed and Open-Circuit Battery Voltage

We started the experiment at 11:14 AM. In order to discharge the battery, we loaded it with the inverter which itself was loaded with an extension cord loaded with two charging laptops, two benchtop DMMs, and 1 lightbulb. Under those conditions, of course the loaded voltage of the battery goes down when discharging and it was 12.3V – an unrealistic metric. When taking data, the extension cord was turned off to get a more realistic unloaded battery voltage.

Time	Energy Consumed (Wh)	Battery Open Circuit Voltage (V)
11:14	0	12.88
11:36	30	12.53
11:50	60	12.45
12:17	100	12.36
12:35	120	12.33

Steps 3 & 7 Battery Energy Data

b. PV Panel Characteristics

i. Characterizing PV Panel IV Curve (steps 4, 5, 6)

For the incremental steps we used to increase the rheostat, the highest power occurred **at $V_{pv} = 16.273V$, $I_{pv} = 4.832A$. This occurred roughly at 12:25 PM. It was 78.63W.** The tabulated data on the following page covers the entire IV curve for shaded and unshaded panel tests.

Test Conditions: UNSHADED, 12:25 PM, Solar Irradiance = 1114 W/m ² <i>*Isc and Voc</i>			Test Conditions: HALF-SHADED, 12:31 PM, Solar Irradiance = 1114 W/m ²		
Vpv (V)	Ipv (A)	Ppv (W)	Vpv (V)	Ipv (A)	Ppv (W)
0.637	5.326	3.392662	0.636	5.436	3.457296
1.005	5.326	5.35263	1.025	5.436	5.5719
2.232	5.317	11.86754	2.879	5.404	15.55812
4.16	5.29	22.0064	5.084	5.33	27.09772
6.551	5.247	34.3731	6.49	5.117	33.20933
9.948	5.23	52.02804	7.111	4.814	34.23235
11.698	5.209	60.93488	7.842	4.092	32.08946
14.76	5.095	75.2022	8.249	3.474	28.65703
16.273	4.832	78.63114	8.505	2.988	25.41294
17.04	4.546	77.46384	8.646	2.701	23.35285
17.882	4.057	72.54727	8.763	2.438	21.36419
18.333	3.693	67.70377	8.866	2.19	19.41654
18.682	3.36	62.77152	8.977	1.919	17.22686
18.952	3.048	57.7657	9.023	1.781	16.06996
19.175	2.78	53.3065	9.088	1.623	14.74982
19.33	2.58	49.8714	9.136	1.493	13.64005
19.49	2.371	46.21079	9.178	1.373	12.60139
19.61	2.204	43.22044	9.208	1.281	11.79545
19.695	2.076	40.88682	9.243	1.187	10.97144
19.76	1.975	39.026	9.27	1.099	10.18773
19.817	1.856	36.78035	9.287	1.033	9.593471
19.854	1.797	35.67764	9.305	0.969	9.016545
19.9	1.716	34.1484	9.313	0.938	8.735594
19.937	1.657	33.03561	9.321	0.894	8.332974
19.953	1.613	32.18419	9.32	0.855	7.9686
19.975	1.565	31.26088	9.332	0.819	7.642908
20.92	0	0	9.342	0.801	7.482942
			9.345	0.782	7.30779
			9.35	0.754	7.0499
			9.347	0.734	6.860698
			9.348	0.7135	6.669798
			19.7	0	0

Recorded Panel Data (MPPs are shaded green, open/short circuit points are in red)

ii. Final Energy Consumed and Open-Circuit Battery Voltage

As per the table in II.a.iii, the final consumed energy from the battery is 120 Wh (only accurate to the nearest 10 Wh due to "Kill-a-Watt" precision). At this point, the open-circuit battery voltage was 12.33V. This measurement was taken at 12:35PM.

c. Charging the Battery

i. Operating Point in Direct Energy Transfer

When charging, the measured battery voltage is **12.91V** at **5.244A** charging. The battery is getting charged at 67.7W. **This is only 86% of the peak power** the panel could supply for the same conditions. Since the open circuit voltage of the battery before charging was 12.33V, and with 5.244A charging it, it rose to 12.91V, we can deduce that the series Thevenin resistance of the battery must be 110.6 m Ω when charging. However, when discharging, the battery was fully charged at 12.88V and it dropped to 12.3V when loaded with 2 DMMs, 1 lightbulb, and 2 laptops through the inverter. This load consumed 100Wh in 1 hr, so it was consuming 100W on average. 100W at 12.88V means an average current from the battery of 7.764A. Thus, when discharging, and assuming the average current to be the initial current of the battery, the Thevenin resistance of the battery is 74.7 m Ω . The discrepancy lies in the fact that the IV curve of batteries are nonlinear generally, and so this Thevenin resistance is a function of state of charge.

ii. Battery Stowed Away and Charging ✓

d. Data Analysis and Modeling

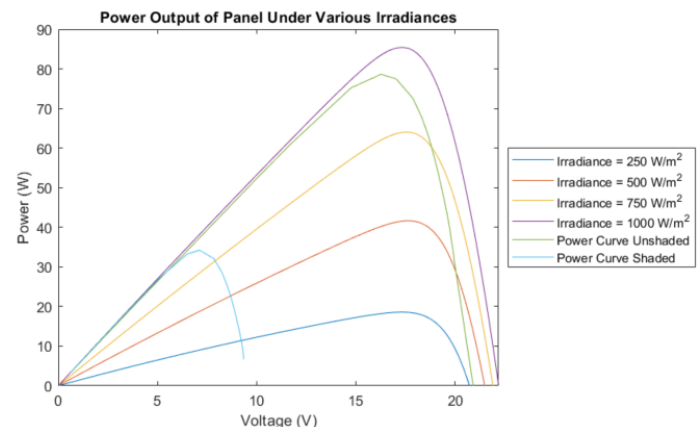
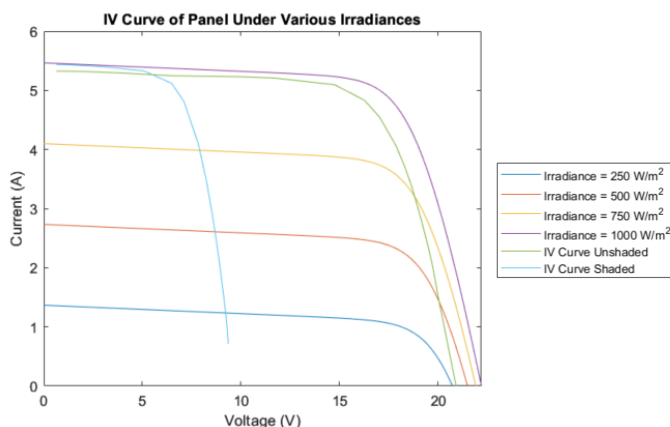
i. Real-World vs Hypothetical Efficiency of Panel

NREL Solar Irradiance at 12:25 PM: 1025.73 W/m², PV Panel Area: 0.633m², Total Incident Power: 649.29W, Generated Power: 78.63W MPP Efficiency: 12.11%

The theoretical efficiency is 13.4%. Our panel appears to have lost some efficiency over the years. We also didn't decrease our step size as we arrived at the MPP, potentially skipping over the actual MPP just because we didn't tabulate it. Additionally, this calculation was done with the NREL-provided irradiance at the date and time of our maximum power point, which was considerably lower than the irradiance we measured. All other calculations/models were done with the measured irradiance of 1114 W/m².

ii. Plotting IV Curves for Normal Operating and Shading vs Model-Predicted Curves from Datasheet Parameters

The data is plotted overlaid on expected IV/Power curves given a k_i , R_p , R_s , and I_{d0} extrapolated from the **datasheet values** for I_{sc} , V_{sc} , I_{mpp} , and V_{mpp} from experiment 1. That is; **$k_i = 0.0054846 \text{ Am}^2/\text{W}$, $I_{D0} = 258.55 \text{ pA}$, $R_p = 1.9323$, and $R_s = 0.4419\Omega$ (12.275m Ω per cell)**

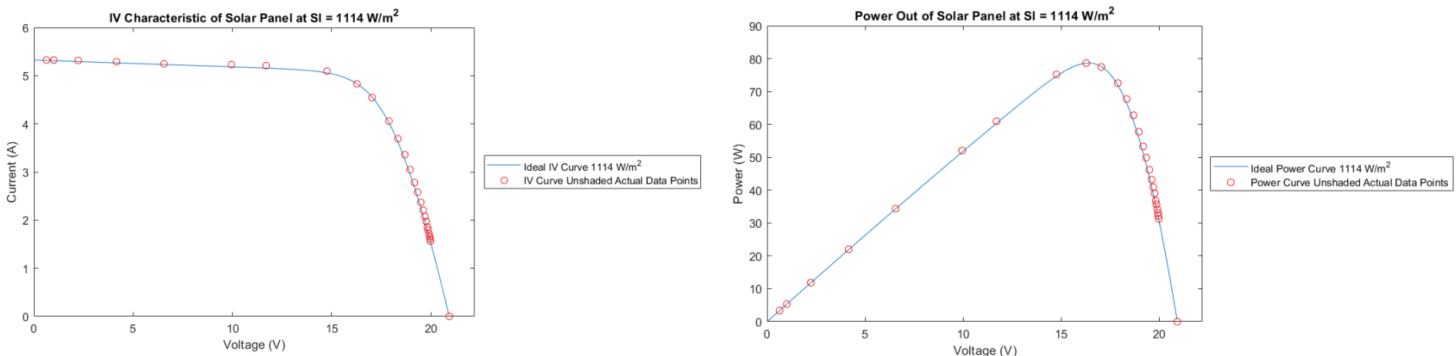


iii. Extrapolating Experimental Values for k_i , ID_0 , R_p , R_s

Clearly, datasheet values lead to a k_i , ID_0 , R_p , and R_s that don't fit our measured data very well at all. To extrapolate experimental panel characteristics using **our own measured I_{sc} , V_{sc} , I_{mpp} , and V_{mpp}** , we reapplied those custom data points to the MATLAB model from Experiment 1. A max power voltage of 16.273 V, current of 4.832 A, irradiance of 1114 W/m², an open circuit voltage of 20.92 V, and a short circuit current of 5.326 A yielded the following results:

K_i	ID_0	R_p	R_s
0.00480754 Am ² /W	993.67 pA	1.9469Ω	0.010839Ω

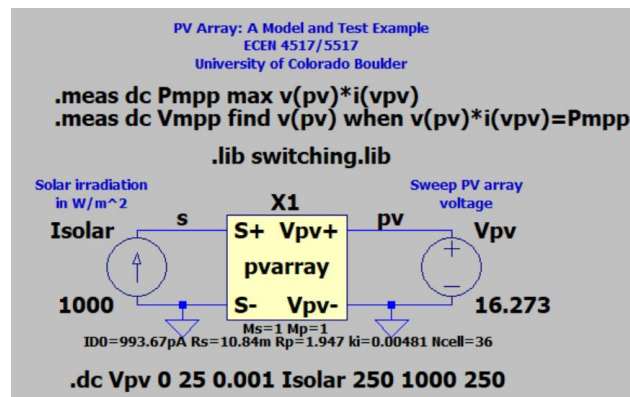
Now, with these renewed, Bench-5-panel-specific values for k_i , ID_0 , R_p , and R_s , the following plot is generated representing what a **model I-V and Power curve should look like at 1114 W/m² compared to the actual data points**.



The model fits the data points extremely well. In fact, originally, instead of red dots, a piece-wise line of the points was overlaid but it was indistinguishable from the curve predicted by the extrapolated values of k_i , ID_0 , R_p , and R_s . Compared to our values determined in Experiment 1, our series contact resistance has gone down slightly, k_i has been reduced, and notably, ID_0 is nearly four times larger than it would've been using the datasheet specifications for V_{oc} , I_{sc} , V_{mpp} , and I_{mpp} . The parallel leakage resistance negligibly changed between the real-world and datasheet models.

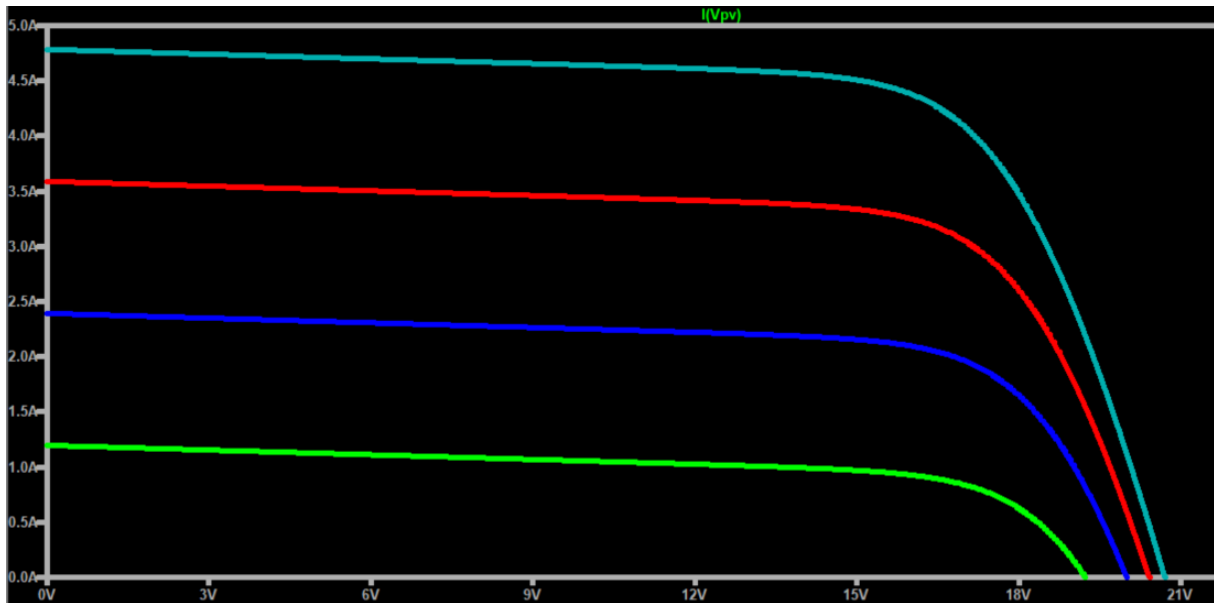
iv. Predicting PV Panel I_{pv} - V_{pv} Characteristics for Different Irradiances

Using the parameter values derived above, we created the following circuit model in LTSpice:

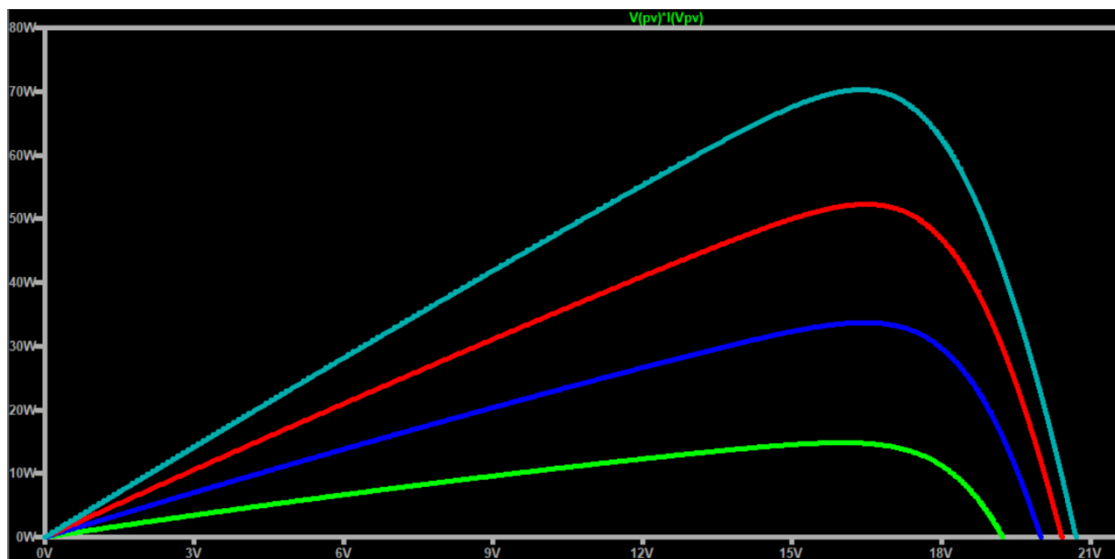


Solar Panel Model Spice Simulation Results

This model predicted the following IV and PV characteristics:

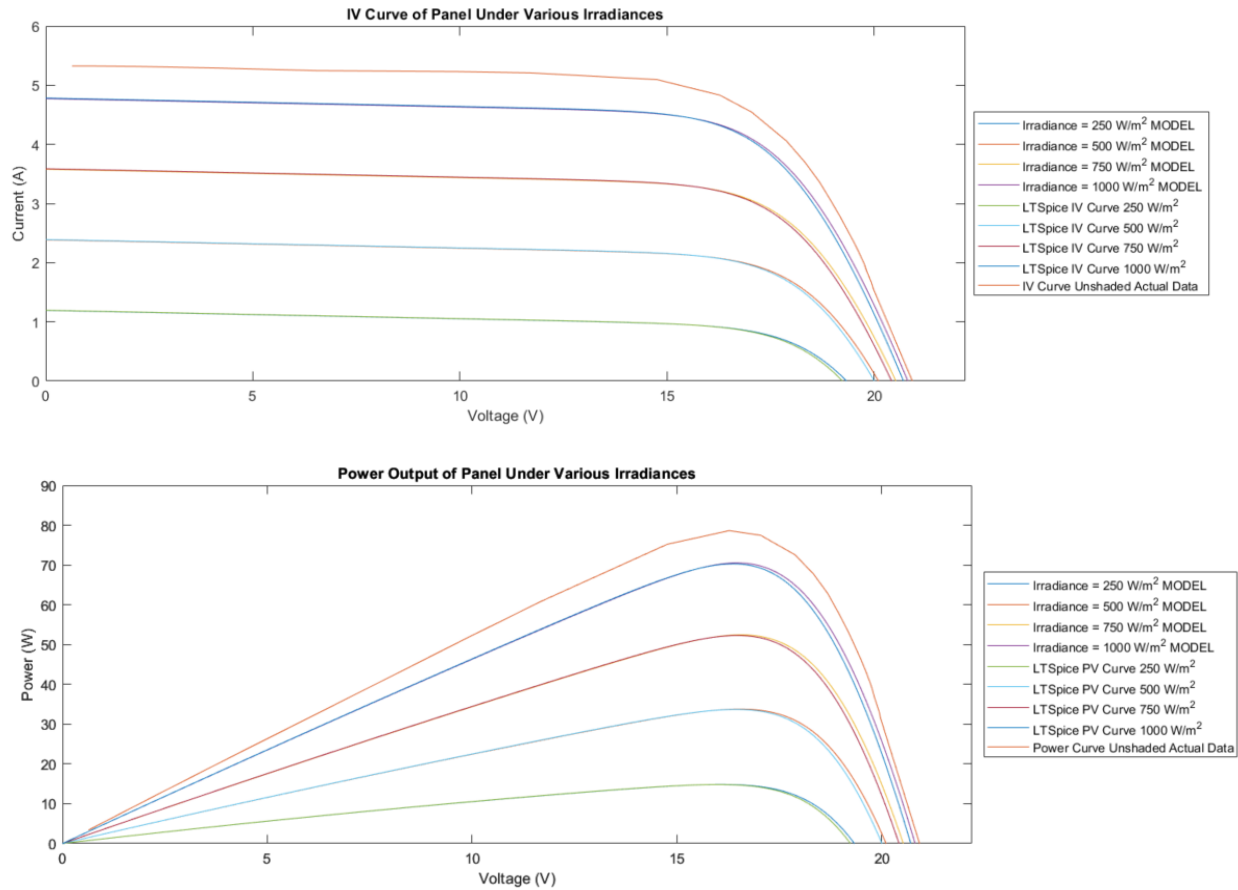


Green: 250 W/m² Blue: 500 W/m² Red: 750 W/m² Cyan: 1000 W/m²



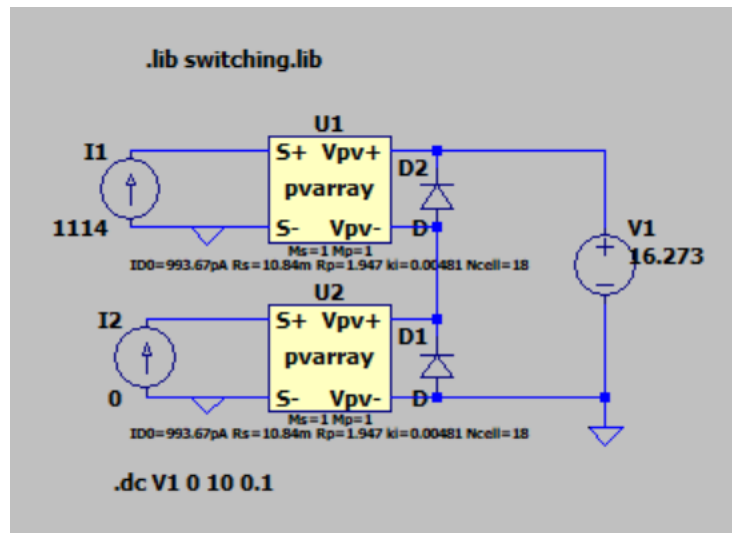
Green: 250 W/m² Blue: 500 W/m² Red: 750 W/m² Cyan: 1000 W/m²

Now, these LTSpice files are overlaid onto similar plots to the ones generated in II.d.ii on the next page. Now, they are compared to the custom MATLAB models for various irradiances. It shows that the MATLAB model and LTSpice models are virtually identical and show that the collected data does indeed have a higher IV and PV curve than the generated curves because that data occurred at an irradiance of 1114 W/m².



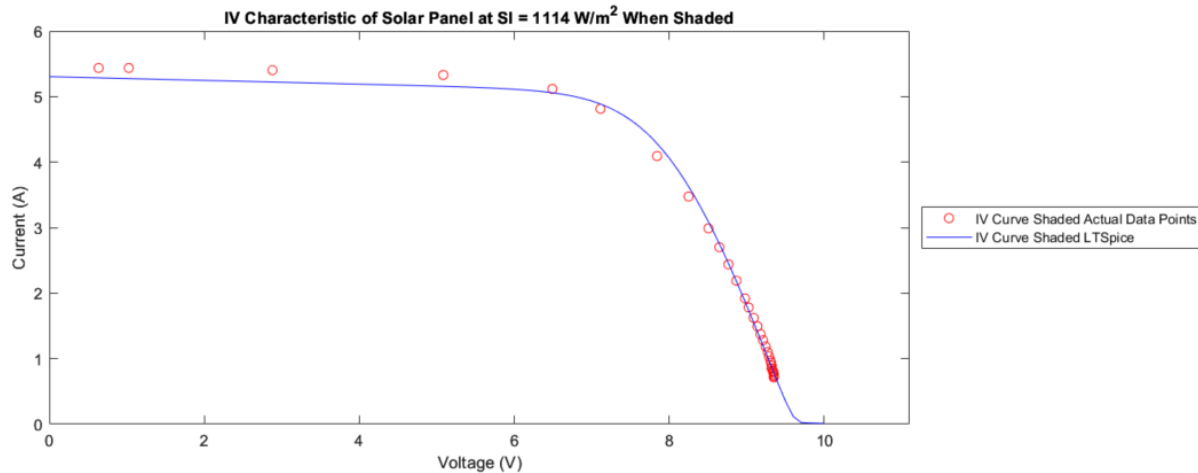
v. Shaded Panel Simulation

To simulate the panel in the shaded condition, the following LTSpice model was created:

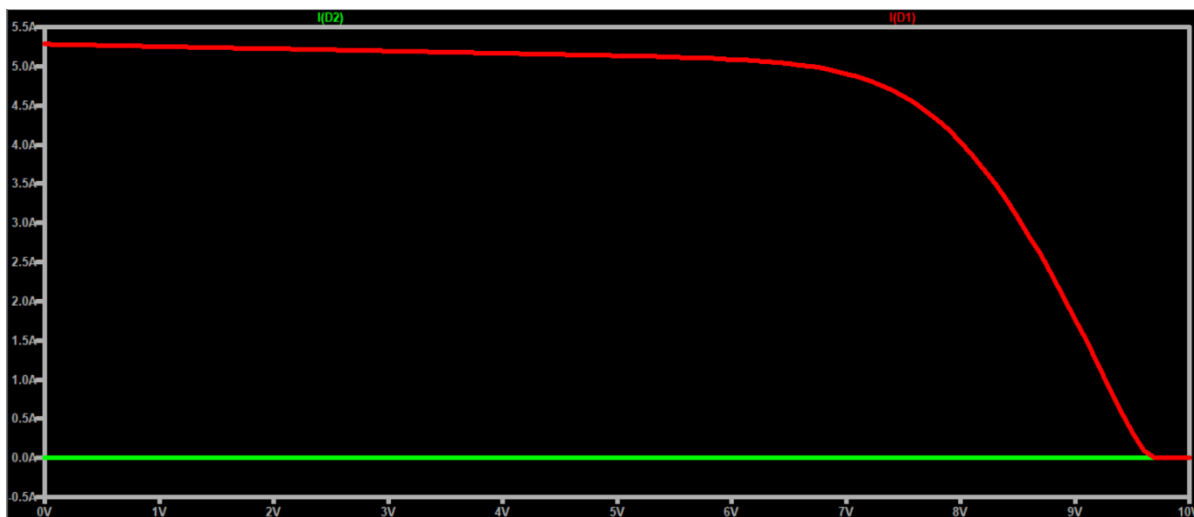


Shaded Panel LTSpice Model

A plot comparing the results generated by this model and our experimental data is shown below:

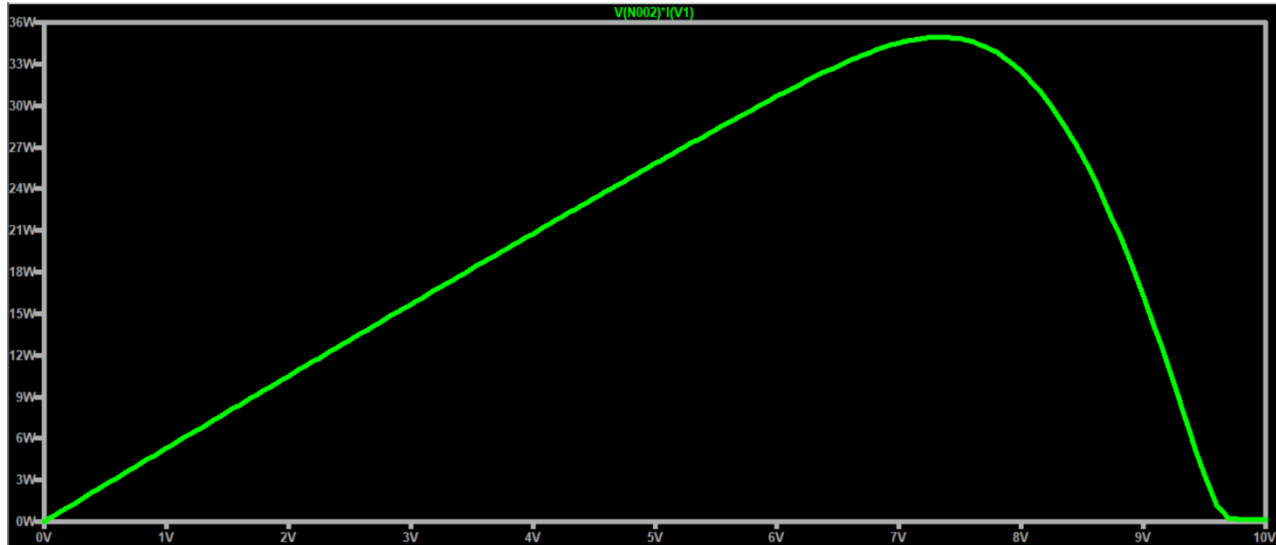


The measured data and the simulated data match extremely well, with notable differences being a slightly lower maximum power in the experimental data. However, the discrepancies are small enough to be attributed to experimental sources of error like parasitic resistance and leakage current. **D1, or the diode in parallel with the string that contained shaded modules, was conducting current as expected and designed for in the shading condition. This is confirmed by LTSpice below.**

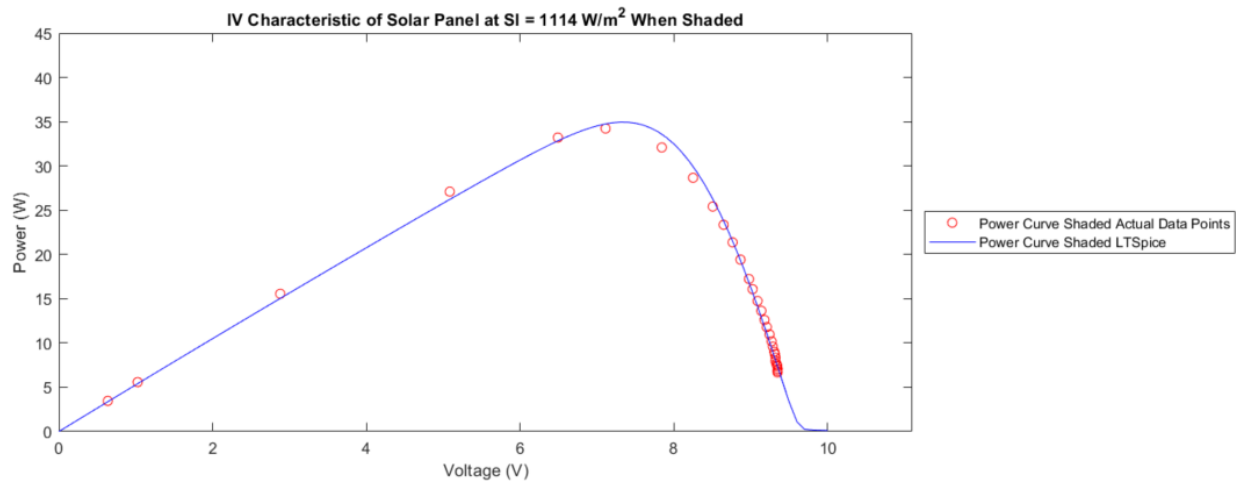


Shaded Model Diode Currents (Green: I(D2), Red: I(D1))

We also simulated and measured shaded PV curves are shown on the following page:



Shaded Model Power Curve



The MPP of the shaded condition occurred at a V_{mpp} of 7.11 V and an I_{mpp} of 4.81 A. This aligns quite well with model-predicted maximum power at 7.31 V and 4.78 A. The LTSpice PV curve fits the actual collected data points remarkably well here too.

vi. Efficiency and Battery Energy Consumption

As shown in the table in II.a.iii, the battery discharged 120 Wh over 81 minutes. We measured the battery's open circuit voltage at several points, and the average voltage over each interval is used to estimate the Amp-hours delivered by the battery. Accounting for the 95% efficiency of the inverter, this estimate gives a total of **10.11 Ah discharged**. The last open circuit battery voltage we measured was 12.33 V, putting the estimated state of charge at about 60% according to the battery SOC table. Since the battery has a capacity of 56 Ah, a 40% discharge would suggest 22.4 Ah were discharged when it went from 12.88 V to 12.33. Our estimated 10.11 Ah would only suggest about 18% discharge. The discrepancy could be accounted for by inaccuracies in our estimate of the voltage of the battery during discharge, since we averaged either end of each interval to find the current despite the characteristic being nonlinear. Another source of error could be with the SOC table predicting charge based on voltage, since our battery started higher than the 12.8 V suggested by the table.

vii. DET Operating Point and MPPT Discussion

As mentioned in II.c.i, the direct energy transfer operating point is 12.91V at 5.244A. **This corresponds to a charging power of 67.7W.** This is only 86% of the possible output power that could be achieved when the panel operates in MPP. By adding a buck converter that ensures the load current (inductor current) is 6.05A at 13V out, the battery will see all of the available 78.6W the panel could offer. It does so primarily via the inductor which reactively stores current from the panel to deliver to the battery even when the panel is disconnected from the inductor. When lossless, power at the load directly translates to power from the panel so designing for a certain power at the load will always allow the panel to operate in MPP. **Indeed, battery current and thus power would go up by roughly 16% by including this buck converter.**

When shaded, the peak power is a little under half. It would be half if the reverse-biased backplane diode was lossless, but in reality, diodes have forward voltages and series resistances and with large currents can be quite lossy. However, currents stay roughly the same in the panel with just the voltage dropping by a little more than half. For the same duty cycle as unshaded operation, the output voltage just drops by half with output current staying the same. The opposite is what needs to happen. The output voltage must stay at 13V despite the input voltage dropping below 13V. **This requires a buck-boost converter, not just a buck converter. When shaded, it boosts the voltage and, in this state, a controller would know that it only needs to send roughly half the I_{mpp} current to the battery.**

viii. MPPT Buck Converter Preliminary Specifications (Using Datasheet I_{sc} , V_{oc} , I_{mpp} , V_{mpp})

- a) Converter Input: Voltage and Current Range:
 - a. Input Voltage can be anything from **13 to 22.2V: Typically 17.2V**
 - i. Cannot fall below the output battery charging voltage of 13V in a buck converter
 - b. Input Current can be **0A to 5.45A: typically 4.95A**
 - i. No load to short circuit current designed for, at peak power will be 4.95A
- b) Converter Output: Voltage and Current Range:
 - a. Output Voltage must be **13V**
 - i. This is a tightly regulated voltage that is what the battery will see when charging.
 - b. Current Range of **0 to 6.54A**
 - i. This depends on how much power is extracted from the panel and the efficiency of the buck converter. Ideally, it cannot extract more than 6.54A and that is what we will try to extract for an output voltage of 13V. ($6.54A \times 13V = 85W = P_{mpp}$)
- c) Converter Switching Frequency
 - a. 100 kHz will be used for its ease in hand and head calculations. Any other specifications reliant on this frequency will be made up through obscure values for other variables like inductance in switching ripple specification.

- d) For $V_{out} = 12V$, $V_{in} = 17V$, if P_{in} is 0 to 85W, peak efficiency depends on switch selection.
 - a. When using diodes in the buck converter, the efficiency will not likely be higher than 90% at peak efficiency and when using synchronous MOSFET switches, expect no more than 96% efficiency. From experience, these efficiencies are quite ubiquitous for this power range and switch configurations.